

Seven configurations. The measured tradeoffs.

Matched recorded-speech comparison · 14 September 2026 · AMD Ryzen 7 PRO 8840HS · CPU only.

MODEL / TESTED CONFIGURATION	NO COMPOSITE SCORE				
	WER %	NORMALIZED WER %	FIRST OUTPUT s · median	CPU WORK s / audio s	PEAK RSS GiB
NVIDIA Nemotron 0.6B Q8	5.33	4.68	1.497	0.939	0.978
Moonshine Small	7.62	6.96	1.965	3.165	0.521
Moonshine Medium	7.94	7.29	1.968	3.753	0.840
Qwen3-ASR 0.6B	4.68	4.46	2.728	2.975	2.864
Qwen3-ASR 1.7B	6.42	5.98	3.710	6.679	6.799
Parakeet TDT v3	7.18	6.53	Unavailable	0.434	3.118
Whisper base.en	9.36	8.71	Unavailable	0.803	0.403

Why Nemotron: early output with practical CPU and memory use.

Qwen 0.6B had lower measured WER. The selection reflects the combined live-dictation tradeoff in this test.

One speaker · 3 previously used recordings · 919 distinct words · 3 repeats · 63 selected successful trials / 64 attempts.

WER excludes punctuation. First output is a recognizer callback, not cursor paint. Offline first output is unavailable.

CPU is not battery use; sampled process-tree RSS may double-count shared pages. No universal superiority is established.

Word error rate

Same recordings. Same references. All seven tested configurations.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Moonshine Small

STREAMING ADAPTER

Moonshine Medium

STREAMING ADAPTER

Qwen3-ASR 0.6B

STREAMING ADAPTER

Qwen3-ASR 1.7B

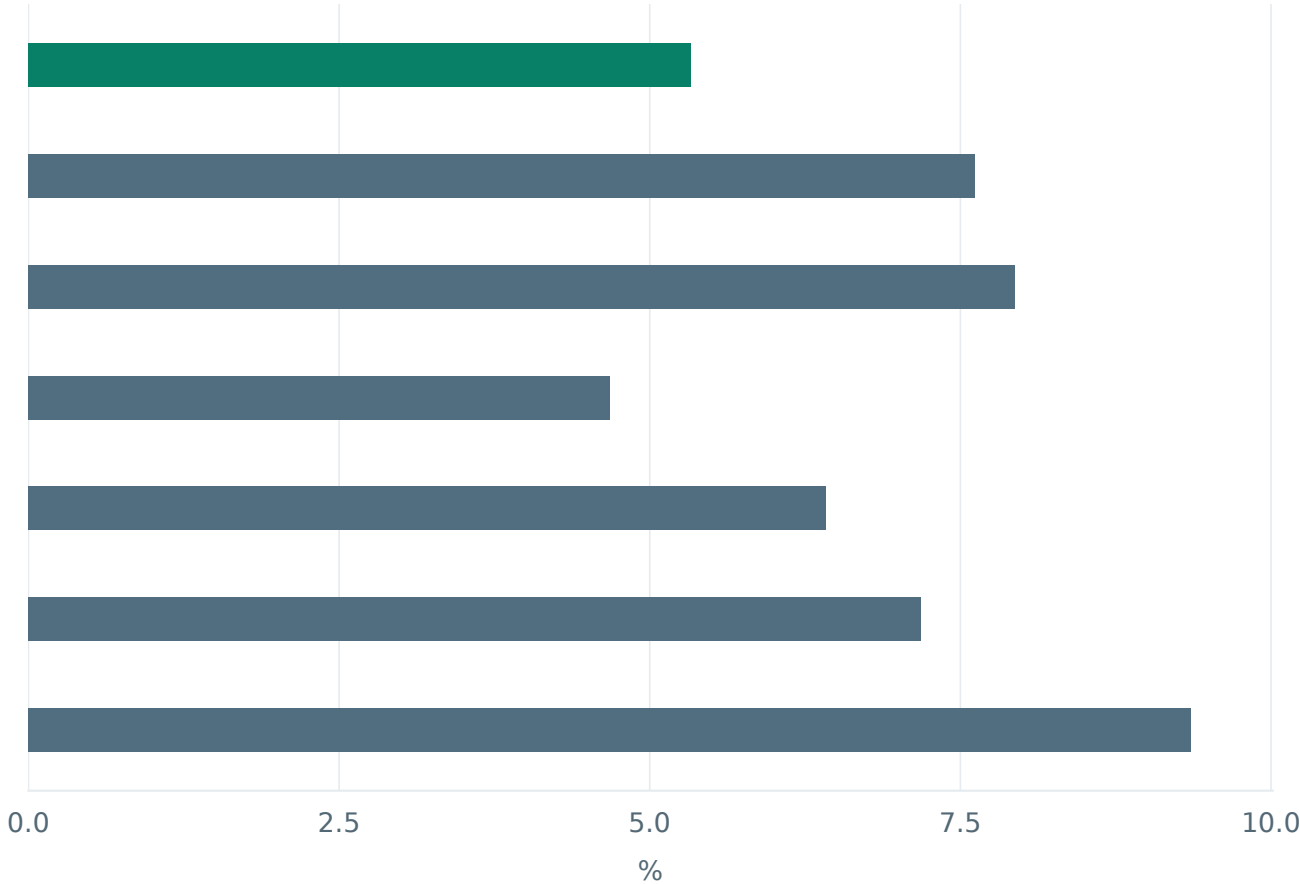
STREAMING ADAPTER

Parakeet TDT v3

OFFLINE ADAPTER

Whisper base.en

OFFLINE ADAPTER



Ignores punctuation and case; number rendering still affects errors.

919 distinct reference positions, replayed three times: 2,757 word observations per model.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Contraction-normalized word error rate

Same recordings. Same references. All seven tested configurations.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Moonshine Small

STREAMING ADAPTER

Moonshine Medium

STREAMING ADAPTER

Qwen3-ASR 0.6B

STREAMING ADAPTER

Qwen3-ASR 1.7B

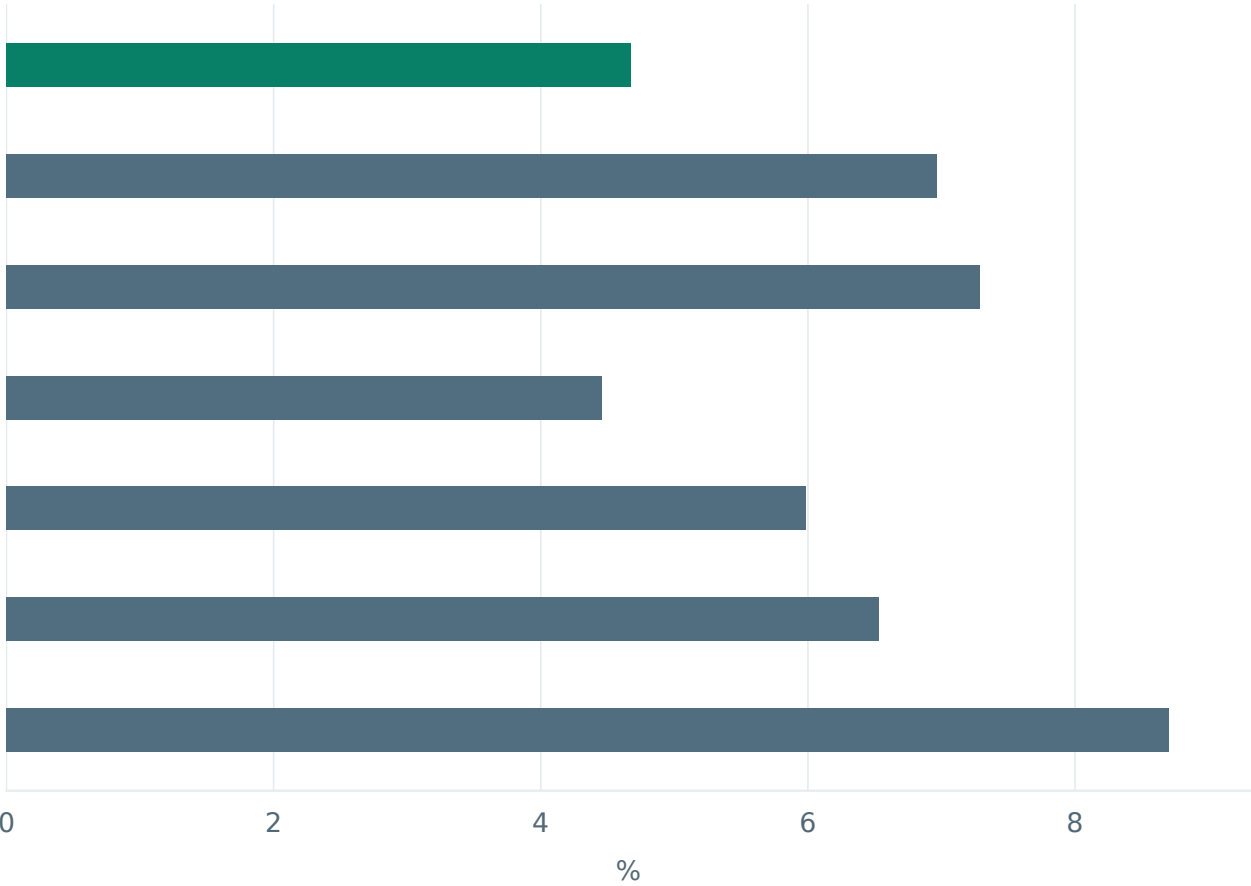
STREAMING ADAPTER

Parakeet TDT v3

OFFLINE ADAPTER

Whisper base.en

OFFLINE ADAPTER



Also expands contractions; this is a separate sensitivity score, not semantic accuracy.

919 distinct reference positions, replayed three times: 2,757 word observations per model.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

First recognizer output

Median from audio sample zero · nine trials per streaming configuration.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

1.497

Moonshine Small

STREAMING ADAPTER

1.965

Moonshine Medium

STREAMING ADAPTER

1.968

Qwen3-ASR 0.6B

STREAMING ADAPTER

2.728

Qwen3-ASR 1.7B

STREAMING ADAPTER

3.710

Parakeet TDT v3

OFFLINE ADAPTER

Not measured · offline adapter

—

Whisper base.en

OFFLINE ADAPTER

Not measured · offline adapter

—



Recognizer callback timing, not first correct word, cursor appearance or pixel paint.

Parakeet and Whisper used offline adapters in this comparison.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

CPU work per second of audio

Median across nine trials · lower values mean less measured CPU work.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

0.939

Moonshine Small

STREAMING ADAPTER

3.165

Moonshine Medium

STREAMING ADAPTER

3.753

Qwen3-ASR 0.6B

STREAMING ADAPTER

2.975

Qwen3-ASR 1.7B

STREAMING ADAPTER

6.679

Parakeet TDT v3

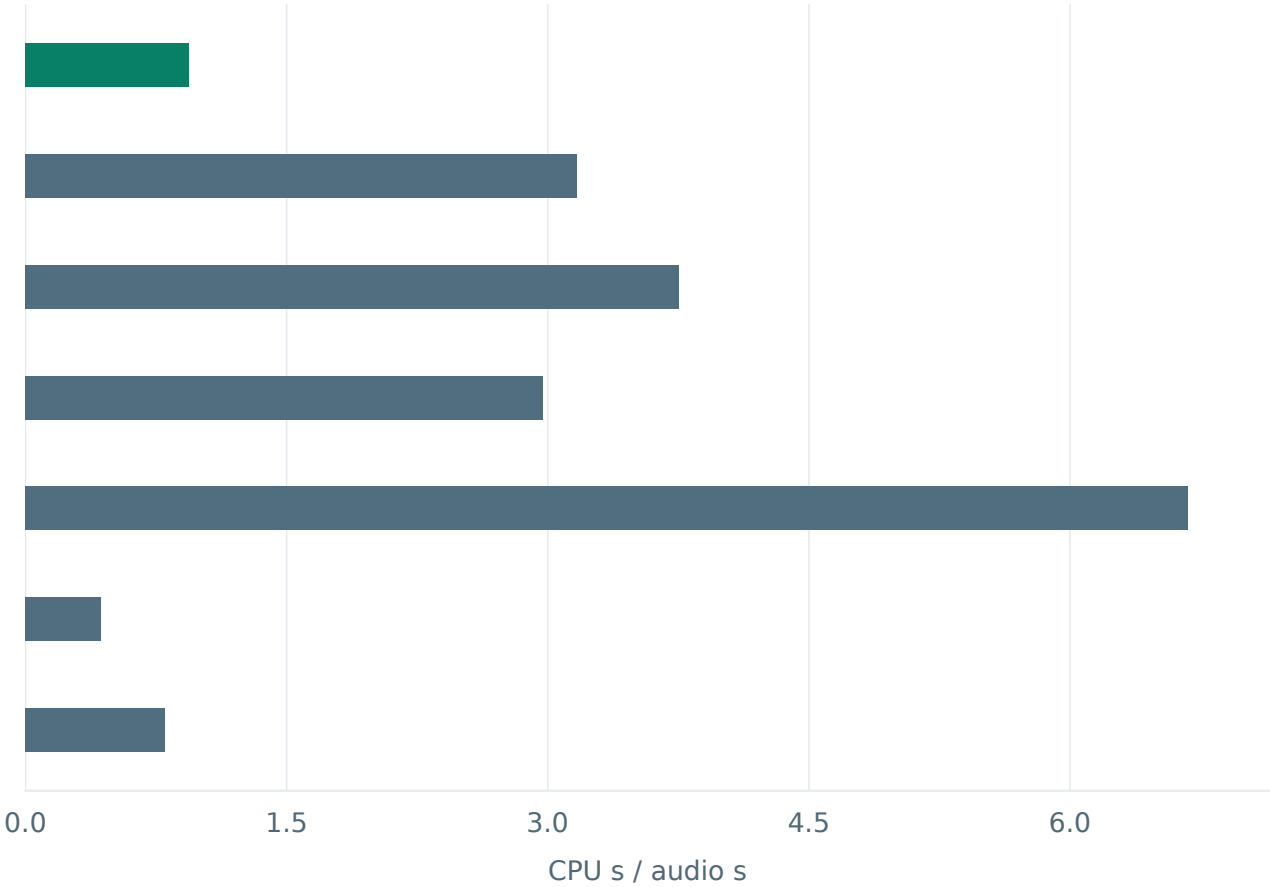
OFFLINE ADAPTER

0.434

Whisper base.en

OFFLINE ADAPTER

0.803



Total CPU work across threads, including replay controller; excludes model load and warmup.

CPU-only execution; runtimes use different thread settings. This is not battery consumption.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Peak sampled resident memory

Maximum across nine trials · process-tree RSS, including startup.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

0.978

Moonshine Small

STREAMING ADAPTER

0.521

Moonshine Medium

STREAMING ADAPTER

0.840

Qwen3-ASR 0.6B

STREAMING ADAPTER

2.864

Qwen3-ASR 1.7B

STREAMING ADAPTER

6.799

Parakeet TDT v3

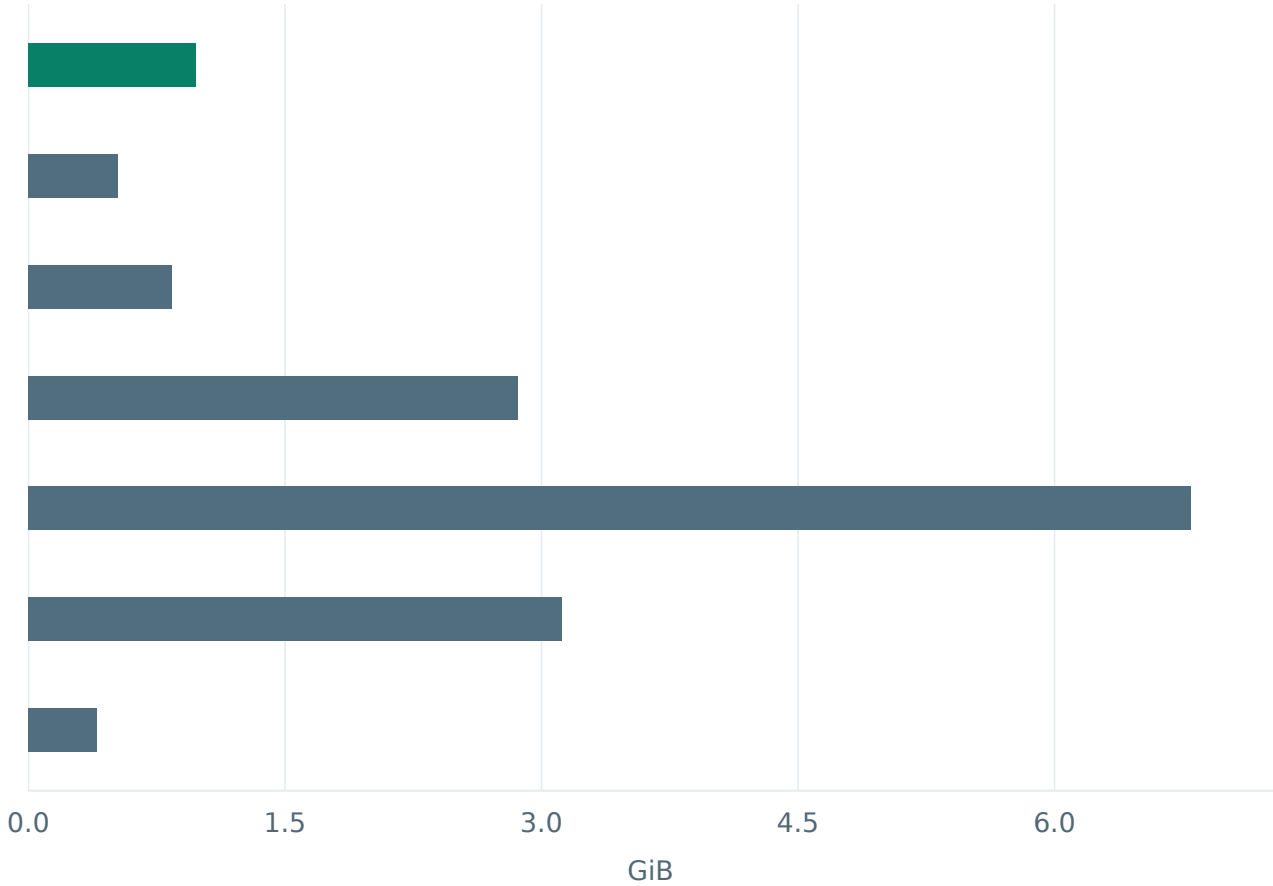
OFFLINE ADAPTER

3.118

Whisper base.en

OFFLINE ADAPTER

0.403



Sampled process-tree sum includes model load and warmup; shared pages may count more than once.

Samples can miss short-lived peaks. This is not whole-system memory use.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Work remaining after speech ends

Same 303.712-second recording · median of three streaming trials.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

89.59 ms

Moonshine Small

STREAMING ADAPTER

1.46 ms

Moonshine Medium

STREAMING ADAPTER

30.80 ms

Qwen3-ASR 0.6B

STREAMING ADAPTER

1.997 s

Qwen3-ASR 1.7B

STREAMING ADAPTER

236.643 s

Parakeet TDT v3

OFFLINE ADAPTER

Not measured · offline adapter

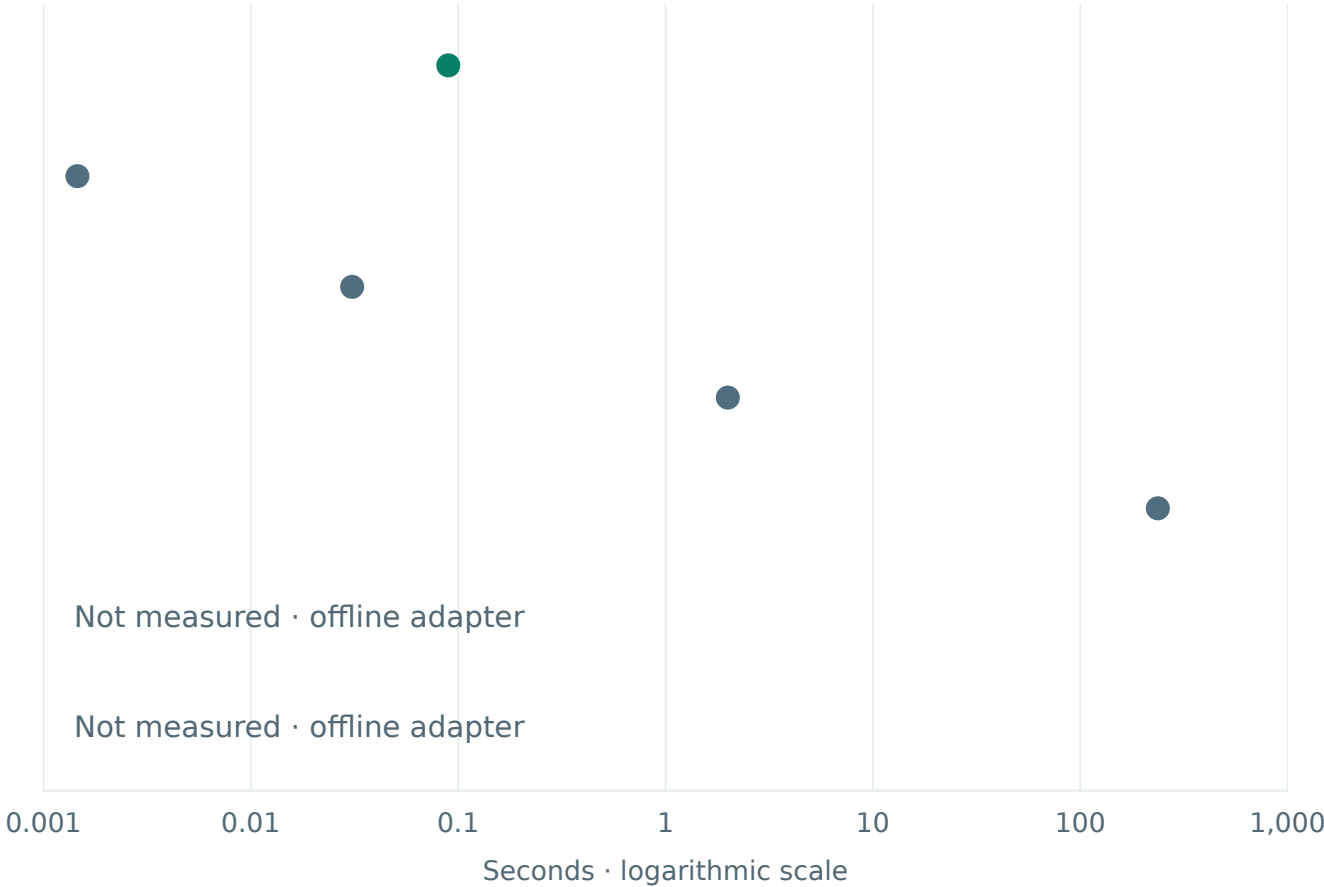
—

Whisper base.en

OFFLINE ADAPTER

Not measured · offline adapter

—



Completion minus scheduled audio end; queued work remains visible. This is not app Stop-to-idle.

Logarithmic seconds axis; labels use ms below one second. Each equal axis step represents a 10× change.

Long-recording slice · one speaker · 303.712 s · 3 repeats per model · drawn from the 63/64-trial comparison.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Text available when audio ends

Same 303.712-second recording · median of three streaming trials.

MODEL / TESTED CONFIGURATION

HIGHER MEANS MORE OUTPUT

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Moonshine Small

STREAMING ADAPTER

Moonshine Medium

STREAMING ADAPTER

Qwen3-ASR 0.6B

STREAMING ADAPTER

Qwen3-ASR 1.7B

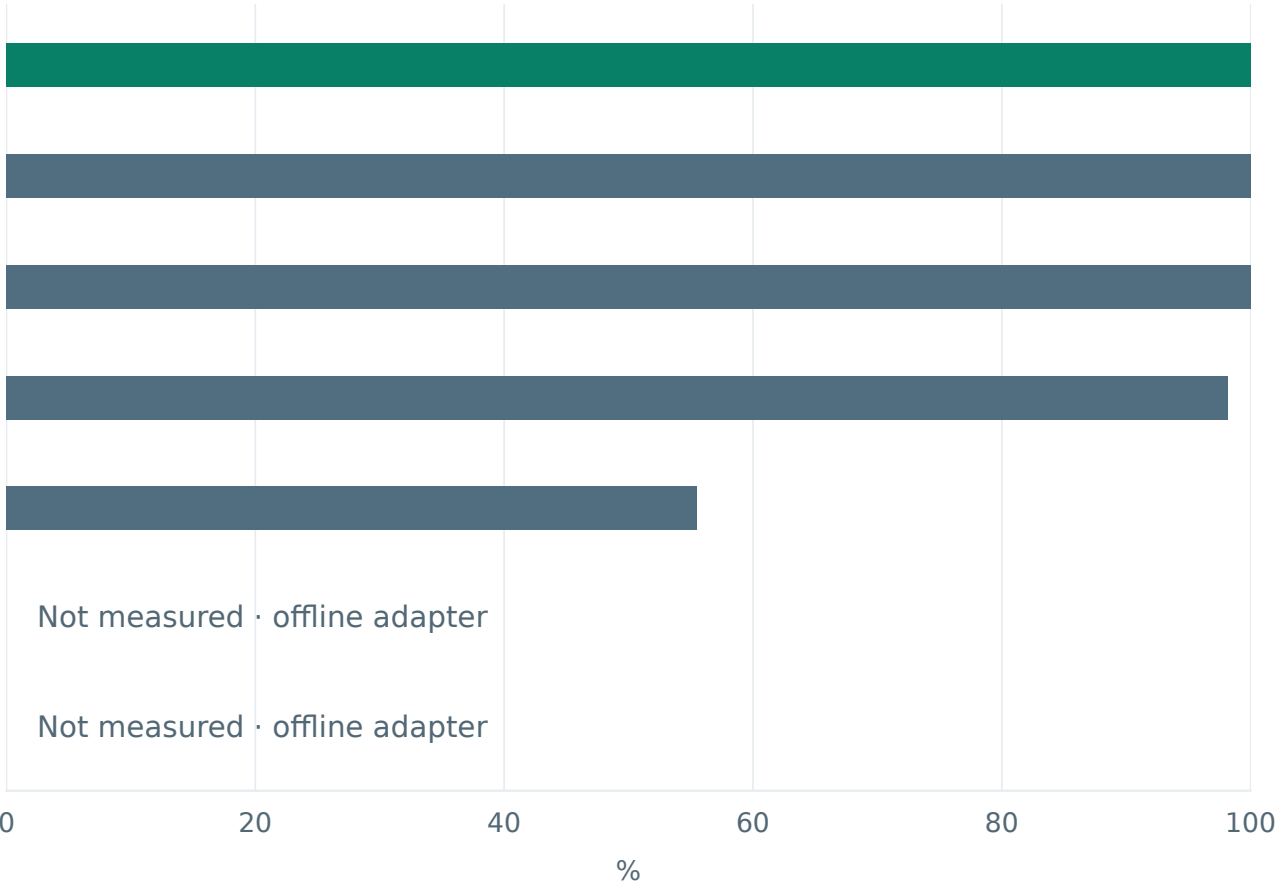
STREAMING ADAPTER

Parakeet TDT v3

OFFLINE ADAPTER

Whisper base.en

OFFLINE ADAPTER



100.00

100.00

100.00

98.14

55.49

—

—

Hypothesis word count / final hypothesis word count. Provisional words may be revised.

This measures output progress, not word accuracy or text committed to another application.

Long-recording slice · one speaker · 303.712 s · 3 repeats per model · drawn from the 63/64-trial comparison.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Completed-prefix revisions

Same 303.712-second recording · median count across three streaming trials.

MODEL / TESTED CONFIGURATION

DESCRIPTIVE · NOT AN ACCURACY SCORE

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Moonshine Small

STREAMING ADAPTER

Moonshine Medium

STREAMING ADAPTER

Qwen3-ASR 0.6B

STREAMING ADAPTER

Qwen3-ASR 1.7B

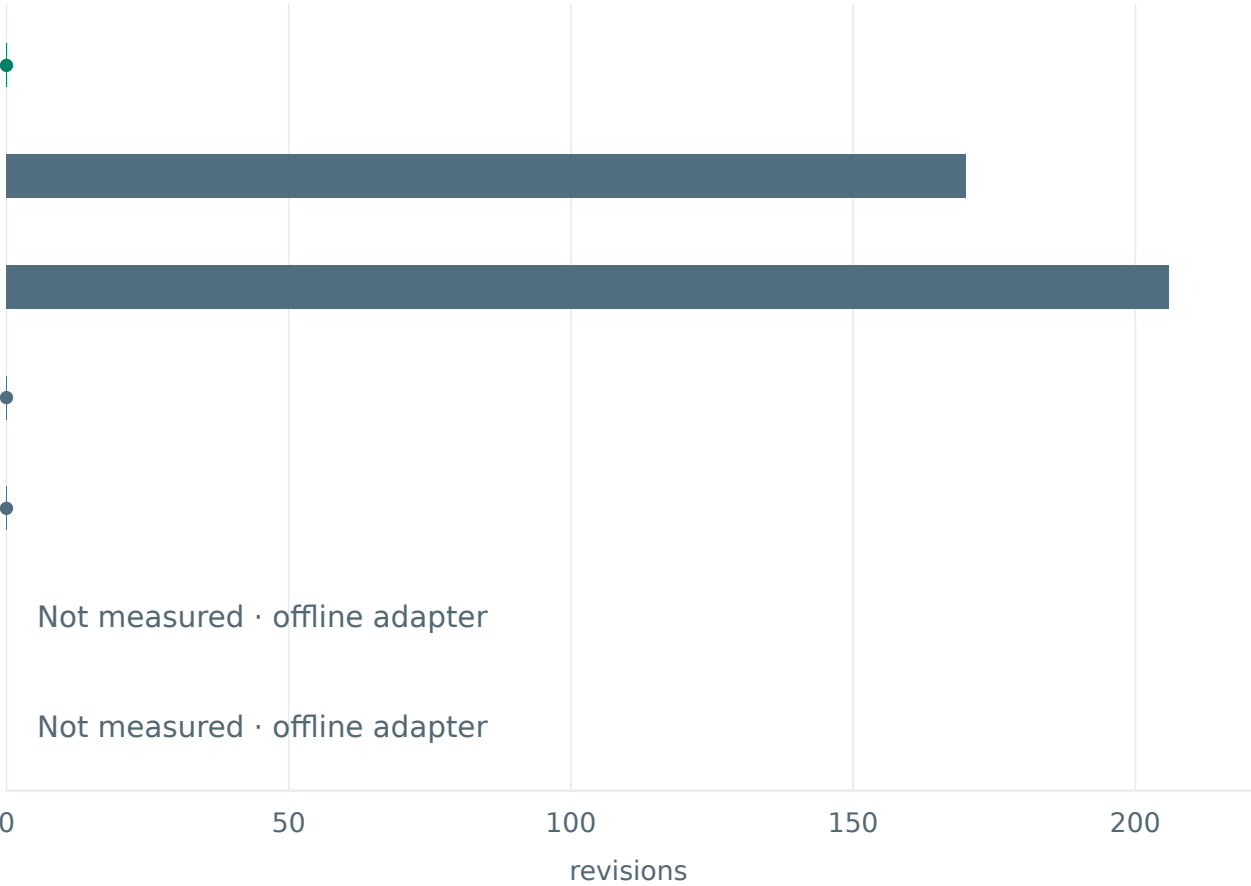
STREAMING ADAPTER

Parakeet TDT v3

OFFLINE ADAPTER

Whisper base.en

OFFLINE ADAPTER



Counts changes to earlier completed hypothesis words; unfinished-word extensions are excluded.

Revisions can be useful corrections. A zero count does not prove that the recognized text is correct.

Long-recording slice · one speaker · 303.712 s · 3 repeats per model · drawn from the 63/64-trial comparison.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Longest observed text-update gap

Same 303.712-second recording · maximum across three streaming trials.

MODEL / TESTED CONFIGURATION

DESCRIPTIVE · INCLUDES SILENCE

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Moonshine Small

STREAMING ADAPTER

Moonshine Medium

STREAMING ADAPTER

Qwen3-ASR 0.6B

STREAMING ADAPTER

Qwen3-ASR 1.7B

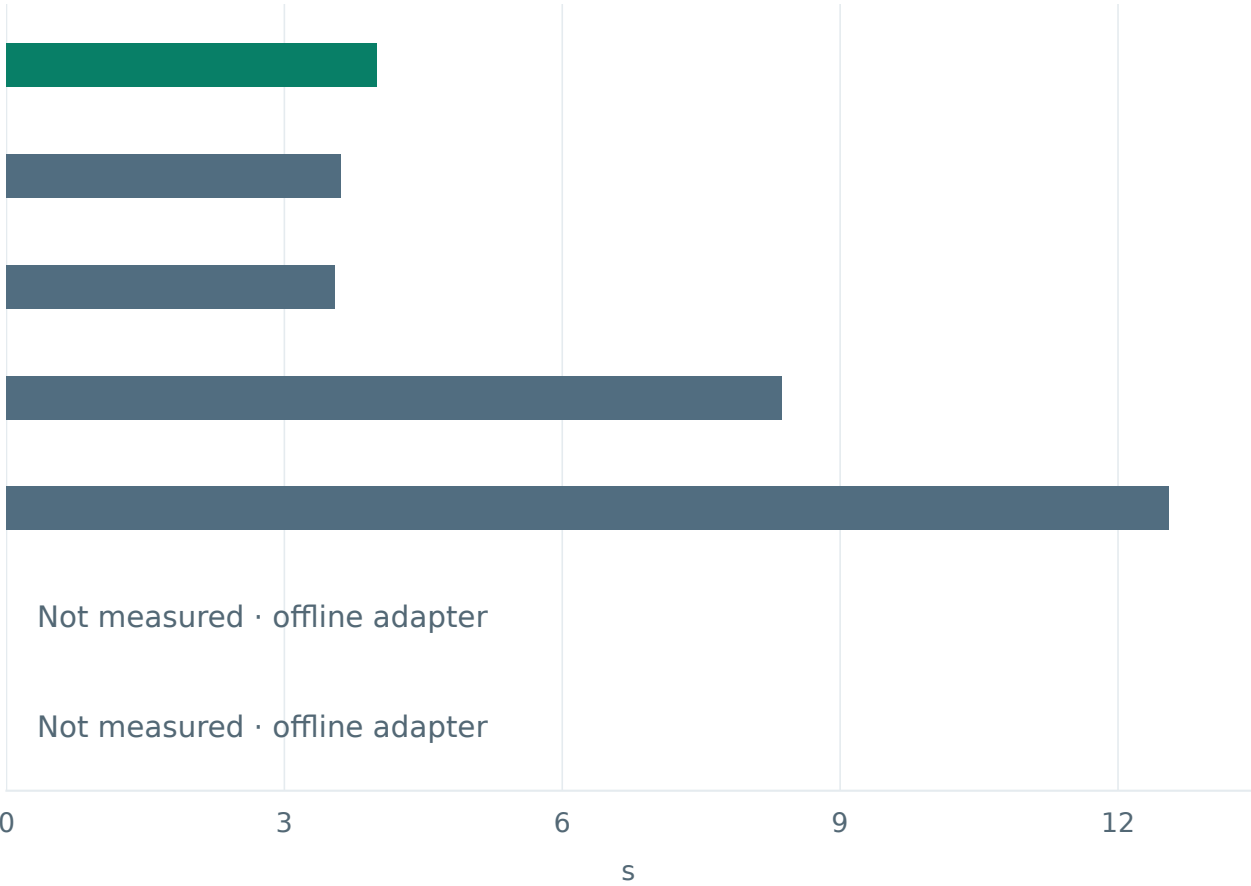
STREAMING ADAPTER

Parakeet TDT v3

OFFLINE ADAPTER

Whisper base.en

OFFLINE ADAPTER



Intervals between recognizer text changes can include legitimate silence and punctuation updates.

This is not voiced-speech lag or a percentile estimate. No acoustic word-end alignment was available.

Long-recording slice · one speaker · 303.712 s · 3 repeats per model · drawn from the 63/64-trial comparison.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Model loading time

Median across nine process launches · retained filesystem cache conditions.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

0.400

Moonshine Small

STREAMING ADAPTER

0.266

Moonshine Medium

STREAMING ADAPTER

0.400

Qwen3-ASR 0.6B

STREAMING ADAPTER

0.476

Qwen3-ASR 1.7B

STREAMING ADAPTER

1.138

Parakeet TDT v3

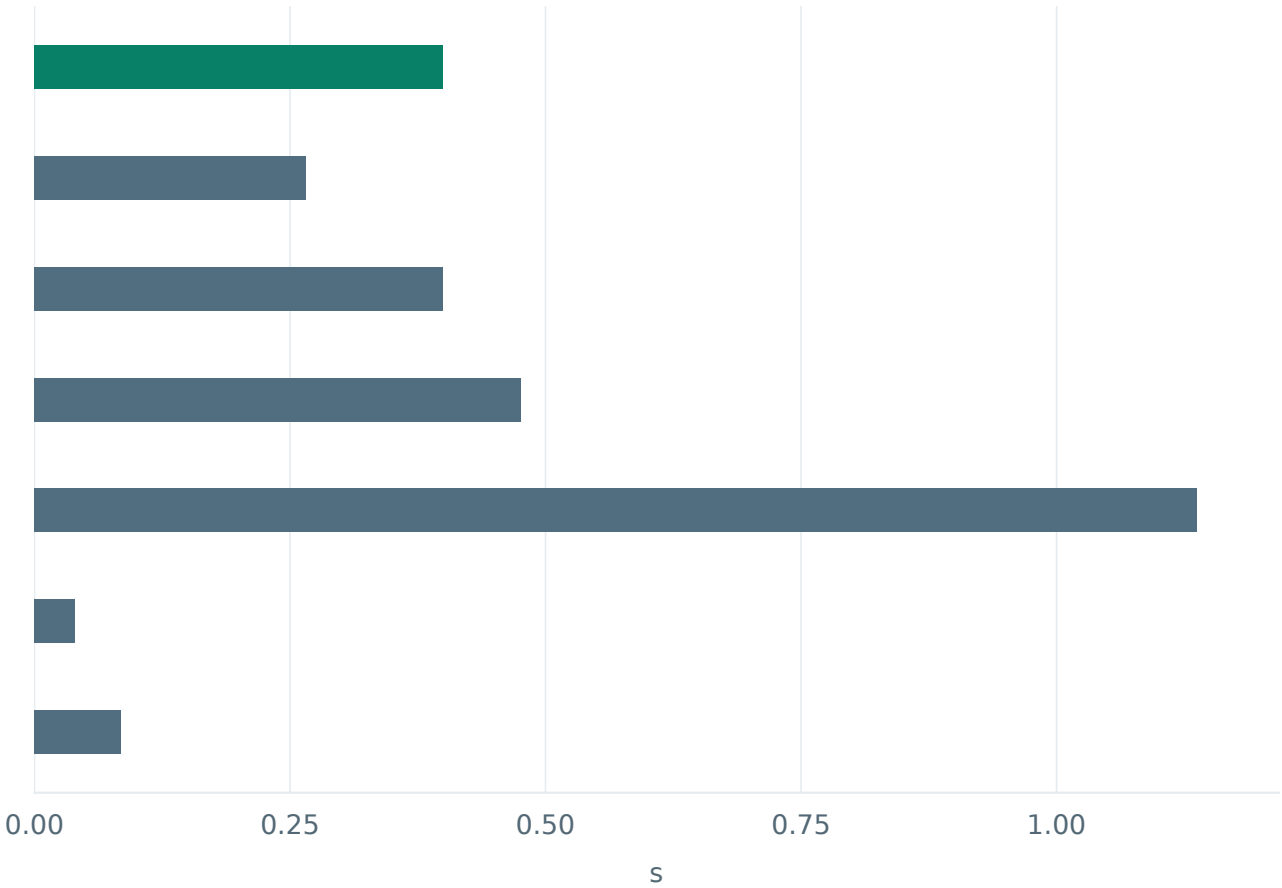
OFFLINE ADAPTER

0.039

Whisper base.en

OFFLINE ADAPTER

0.085



Adapter-reported loading boundary; runtime initialization is not identical across implementations.

OS caches were uncontrolled. This is not cold-boot startup or shortcut-to-ready latency.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Warmup decoding time

Median across nine trials · same separate 2.09-second public warmup.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

1.009

Moonshine Small

STREAMING ADAPTER

0.662

Moonshine Medium

STREAMING ADAPTER

0.695

Qwen3-ASR 0.6B

STREAMING ADAPTER

0.761

Qwen3-ASR 1.7B

STREAMING ADAPTER

1.854

Parakeet TDT v3

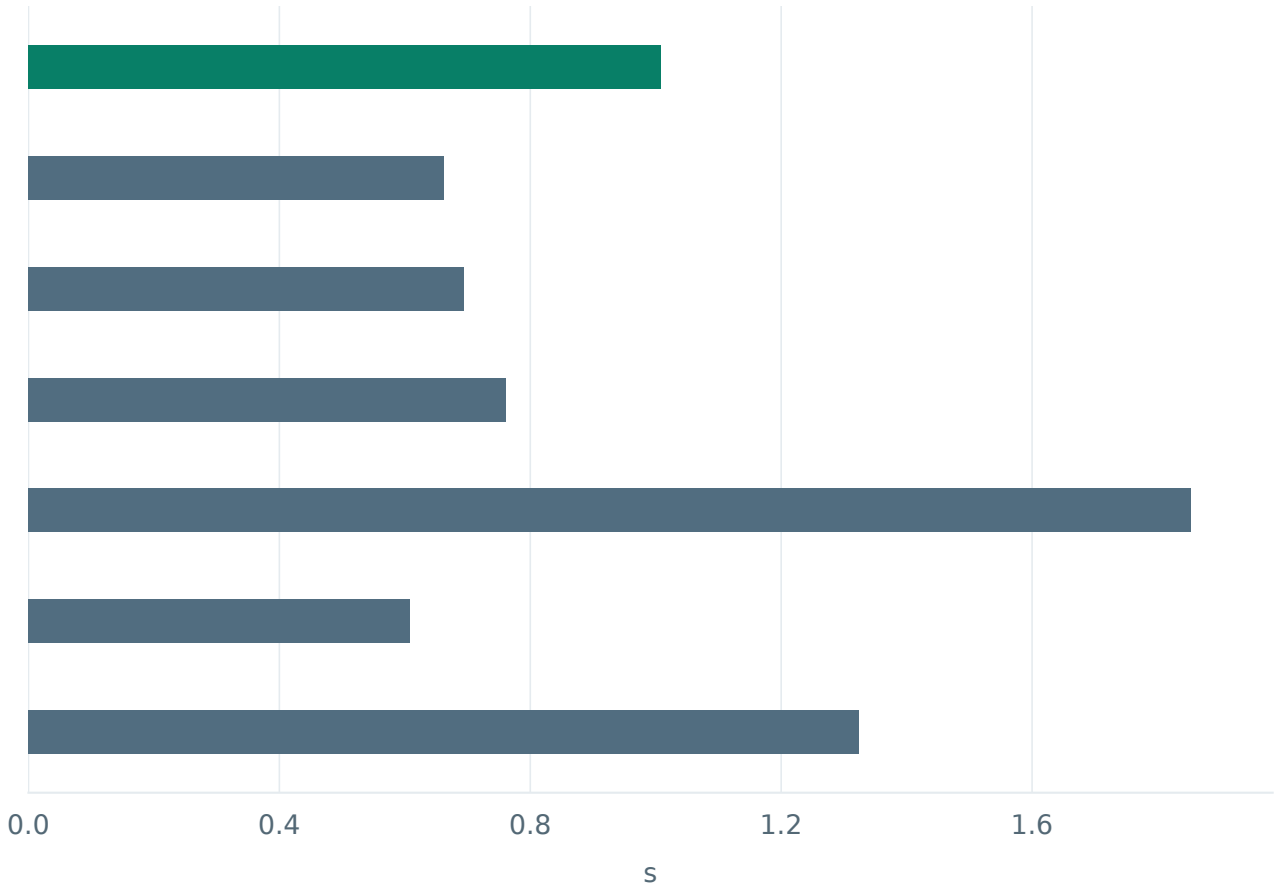
OFFLINE ADAPTER

0.608

Whisper base.en

OFFLINE ADAPTER

1.325



Adapter-reported warmup; separate from model load and the measured comparison recordings.

These runtime boundaries do not establish end-to-end application readiness.

One speaker · 3 previously used recordings · 919 distinct reference words · 63 selected successes / 64 attempts.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Full-recording offline decoding

Same 303.712-second recording · median of three offline trials.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · STREAMING ADAPTER

Not measured · streaming adapter

—

Moonshine Small

STREAMING ADAPTER

Not measured · streaming adapter

—

Moonshine Medium

STREAMING ADAPTER

Not measured · streaming adapter

—

Qwen3-ASR 0.6B

STREAMING ADAPTER

Not measured · streaming adapter

—

Qwen3-ASR 1.7B

STREAMING ADAPTER

Not measured · streaming adapter

—

Parakeet TDT v3

OFFLINE ADAPTER



58.692

Whisper base.en

OFFLINE ADAPTER



39.917

0 15 30 45 60
s

Entire recording decoded offline by the tested Parakeet and Whisper adapters.

Streaming completion tails are a different metric and are not inserted into this comparison.

Long-recording slice · one speaker · 303.712 s · 3 repeats per model · drawn from the 63/64-trial comparison.

14 Sep 2026 · AMD Ryzen 7 PRO 8840HS · CPU only · runtime thread settings differ · no general model ranking.

Earlier public-speech accuracy

Separate cohort · 40 public speakers · 998 reference words · four integrations.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · INCREMENTAL

1.30

Moonshine Small

INCREMENTAL

2.71

Moonshine Medium

INCREMENTAL

1.90

Qwen3-ASR 0.6B

NO TRIAL

Not tested in this cohort

—

Qwen3-ASR 1.7B

NO TRIAL

Not tested in this cohort

—

Parakeet TDT v3

NO TRIAL

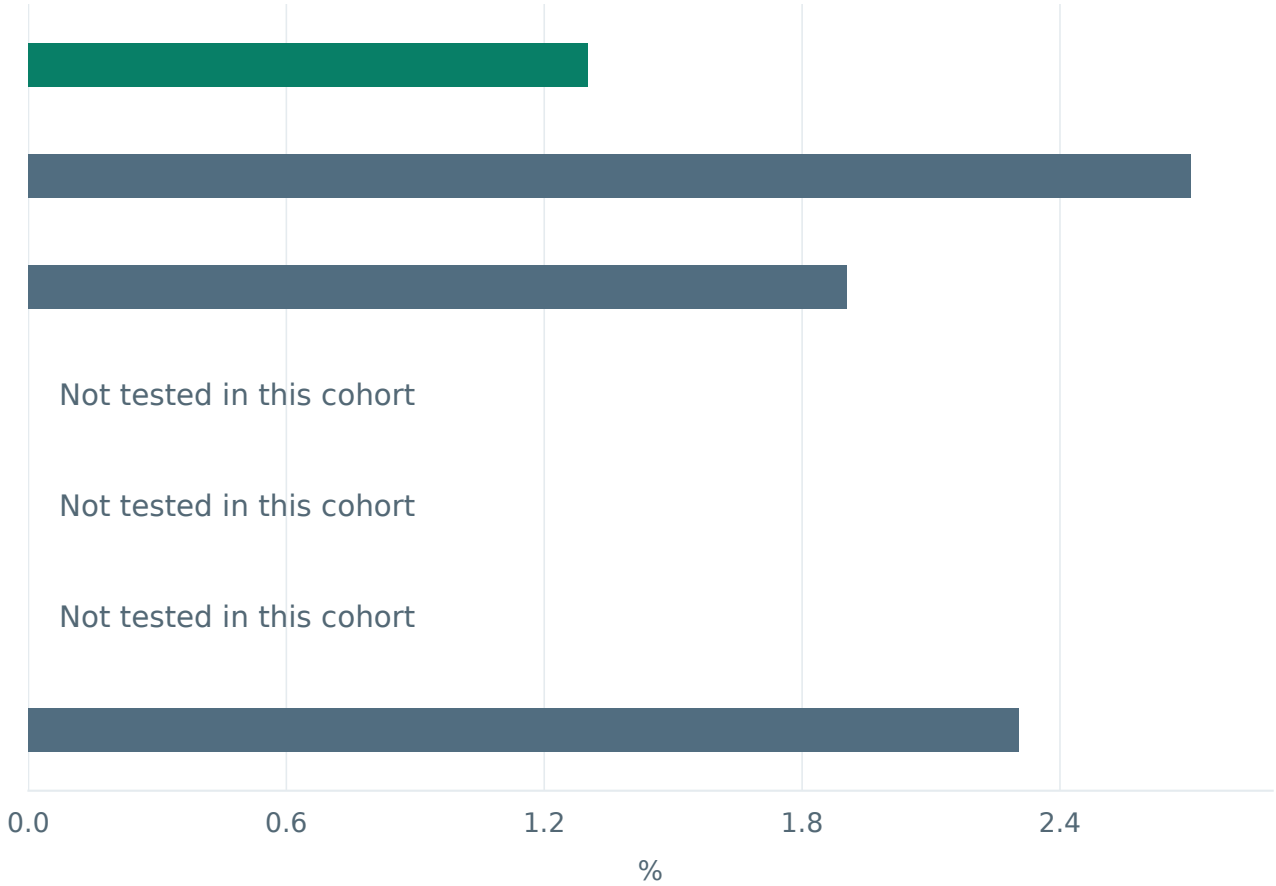
Not tested in this cohort

—

Whisper base.en

FULL UTTERANCE

2.30



Do not combine these rates with the one-speaker recorded-speech comparison. WER excludes punctuation.

Previously evaluated corpus; the NVIDIA-Moonshine Medium accuracy difference is not conclusive.

14 Sep 2026 · 160 completed inferences · CPU only · NVIDIA 13, Small 27, Medium 19, Whisper 23 word errors.

Historical regression corpus; NVIDIA vs Medium descriptive 95% WER-difference interval: −0.11 to +1.41 pp.

Earlier application Stop-to-idle

Separate application test · isolated X11 / GTK · one 47-word public clip.

MODEL / TESTED CONFIGURATION

LOWER IS BETTER

NVIDIA Nemotron 0.6B Q8

SELECTED DEFAULT · VOCO .34

165

Moonshine Small

RESEARCH PROTOTYPE

Incomplete · prototype trial failed

—

Moonshine Medium

RESEARCH PROTOTYPE

Incomplete · prototype trial failed

—

Qwen3-ASR 0.6B

NO TRIAL

Not tested in this cohort

—

Qwen3-ASR 1.7B

NO TRIAL

Not tested in this cohort

—

Parakeet TDT v3

NO TRIAL

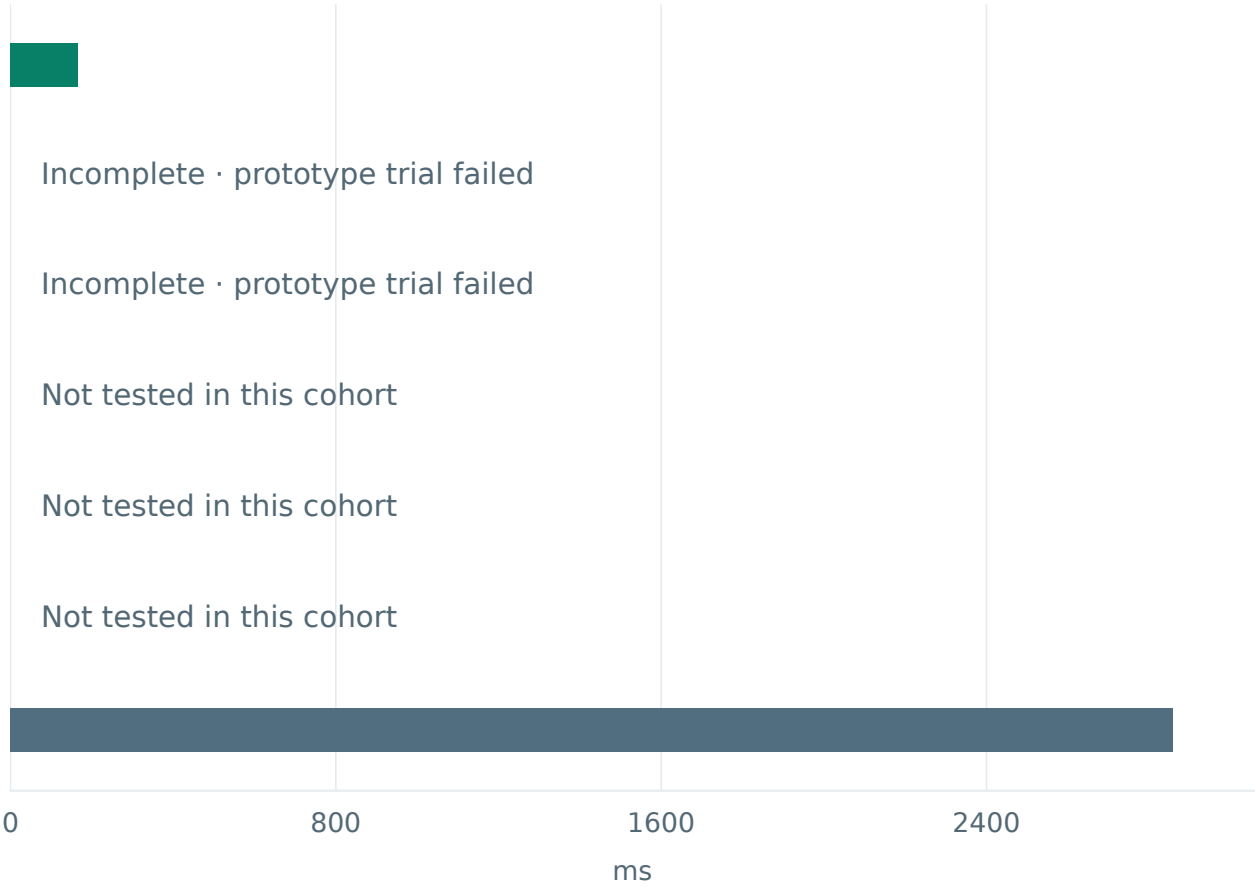
Not tested in this cohort

—

Whisper base.en

VOCO .33

2,859



NVIDIA / VOCO .34 and Whisper / .33: three completed trials each. Different complete app integrations.

Moonshine: one unsuccessful prototype trial per model; no successful Stop time. Not release .43 timings.

14 Sep 2026 · 6 completed trials + 2 failed feasibility trials · CPU only · field/state observation, not pixel paint.

Observed Stop ranges: NVIDIA 161–165 ms; Whisper 2,579–3,066 ms. Small sample; no percentile guarantee.